



SPRING END SEMESTER EXAMINATION-2019

6th Semester B.Tech (Open Elective-I)

DATA STRUCTURE USING C

CS-3040

(For 2017(L.E), 2016 & 2015 Admitted Batches)

Time: 3 Hours

Full Marks: 60

Answer any SIX questions.

Question paper consists of four sections-A, B, C, D.

Section A is compulsory.

Attempt minimum one question each from Sections B, C, D.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1.

[2 × 10]

- (a) What is an ADT?
- (b) What are the different ways to represent a stack?
- (c) Write the overflow and underflow condition of a Circular Queue considering the implementation in an array.
- (d) Construct a binary search tree for the following sequence of numbers.
45,36,76,23,89,115,98,39,41,56,69,48
- (e) Consider an array A[20][10]. Assume 4 bytes per element and the base address of array A is 100. Find the address of A[11][5] assuming row major storage.
- (f) What is an AVL tree? Why it is used?
- (g) Write the C function for push and pop operations of a stack using array.

- (h) Compare between single linked list and double linked list.
- (i) What is B-tree and B+ tree?
- (j) What is a generic pointer? Give one example.

SECTION-B

- 2. (a) Convert the following infix expression to postfix expression using Stack. [4]
$$a+b*(c^d-e)^{(f+g*h)}-i$$
- (b) Write an algorithm/ C function that will divide an array in such a way that all the odd elements will come before all the even numbers. [4]
- 3. (a) What is sparse matrix? How to represent sparse matrix in 3-tuple format? Write the pseudo code to add two sparse matrices. [4]
- (b) Write the enqueue and dequeue operation of circular queue. [4]

SECTION-C

- 4. (a) Write the pseudo code for merge sort. [4]
- (b) Evaluate the following postfix expression using stack. [4]
$$12,7,3,-,/2,1,5,+,*,+$$
- 5. (a) What is Deque? Write algorithm to implement an input restricted deque. [4]
- (b) Write an algorithm /Pseudo code/C code to remove duplicate elements from unsorted single linked list. [4]

6. (a) Sort the following numbers using quick sort. Show all the steps. [4]

60,40,20,95,90,30,50,70

- (b) Construct the binary tree of the following preorder and inorder sequence of traversal given. Also write the postorder traversal of the binary tree. [4]

Preorder: A,B,D,H,I,E,J,K,C,F,G

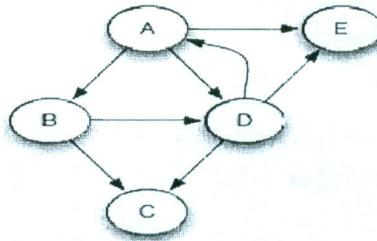
In-order: H,D,I,B,J,E,K,A,F,C,G

SECTION-D

7. (a) Construct an AVL tree with the following set of numbers. [4]

55,66,77,15,11,33,22,35,25

- (b) Design with diagram to traverse the paths in the following given graph using BFS and DFS method starting from node A. [4]



8. (a) Design the insertion and deletion function of the queue which is implemented by using stacks. [4]
- (b) Design an algorithm/C code snippet to delete a node of a binary search tree. [4]
